BANKING SYSTEM

PROGRAM

```
SET SERVEROUTPUT ON;
SET LINESIZE 120;
SET PAGESIZE 50;

BEGIN
   EXECUTE IMMEDIATE 'DROP TABLE Loan_Payment CASCADE CONSTRAINTS';
EXCEPTION
   WHEN OTHERS THEN NULL;
END;
/

BEGIN
   EXECUTE IMMEDIATE 'DROP TABLE Bank_Transaction CASCADE CONSTRAINTS';
EXCEPTION
   WHEN OTHERS THEN NULL;
END;
/

BEGIN
   EXECUTE IMMEDIATE 'DROP TABLE Loan CASCADE CONSTRAINTS';
EXCEPTION
   WHEN OTHERS THEN NULL;
END;
/

BEGIN
   EXECUTE IMMEDIATE 'DROP TABLE Account CASCADE CONSTRAINTS';
EXCEPTION
   WHEN OTHERS THEN NULL;
END;
/

BEGIN
   EXECUTE IMMEDIATE 'DROP TABLE Branch CASCADE CONSTRAINTS';
EXCEPTION
   WHEN OTHERS THEN NULL;
END;
```

```
/

BEGIN
   EXECUTE IMMEDIATE 'DROP TABLE Customer CASCADE CONSTRAINTS';
EXCEPTION
   WHEN OTHERS THEN NULL;
END;
/

CREATE TABLE Customer (
   Customer_ID NUMBER PRIMARY KEY,
   Customer_Name VARCHAR2(100) NOT NULL,
   Date_Of_Birth DATE,
   Gender VARCHAR2(10),
   Phone VARCHAR2(15),
   Email VARCHAR2(100),
   Address VARCHAR2(200)
);

CREATE TABLE Branch (
   Branch_ID NUMBER PRIMARY KEY,
   Branch_Name VARCHAR2(100) NOT NULL,
   City VARCHAR2(50),
   IFSC_Code VARCHAR2(20) UNIQUE
);

CREATE TABLE Account (
   Account_No NUMBER PRIMARY KEY,
   Customer_ID NUMBER,
   Branch_ID NUMBER,
   Account_Type VARCHAR2(30),
   Opening_Date DATE,
   Balance NUMBER(12,2),
   Account_Status VARCHAR2(20),
   CONSTRAINT fk_account_customer
      FOREIGN KEY (Customer_ID)
      REFERENCES Customer(Customer_ID),
   CONSTRAINT fk_account_branch
      FOREIGN KEY (Branch_ID)
      REFERENCES Branch(Branch_ID)
```

```
);

CREATE TABLE Bank_Transaction (
    Transaction_ID NUMBER PRIMARY KEY,
    Account_No NUMBER,
    Transaction_Date DATE,
    Transaction_Type VARCHAR2(20),
    Amount NUMBER(12,2),
    Description VARCHAR2(200),
    CONSTRAINT fk_transaction_account
        FOREIGN KEY (Account_No)
        REFERENCES Account(Account_No)
);

CREATE TABLE Loan (
    Loan_ID NUMBER PRIMARY KEY,
    Customer_ID NUMBER,
    Loan_Type VARCHAR2(50),
    Loan_Amount NUMBER(12,2),
    Interest_Rate NUMBER(5,2),
    Loan_Date DATE,
    Loan_Status VARCHAR2(20),
    CONSTRAINT fk_loan_customer
        FOREIGN KEY (Customer_ID)
        REFERENCES Customer(Customer_ID)
);

CREATE TABLE Loan_Payment (
    Payment_ID NUMBER PRIMARY KEY,
    Loan_ID NUMBER,
    Payment_Date DATE,
    Payment_Amount NUMBER(12,2),
    Payment_Mode VARCHAR2(30),
    CONSTRAINT fk_loan_payment
        FOREIGN KEY (Loan_ID)
        REFERENCES Loan(Loan_ID)
);

INSERT INTO Branch VALUES
(1, 'Chennai Main Branch', 'Chennai', 'BANK000001');
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```
INSERT INTO Branch VALUES
(2, 'Coimbatore Branch', 'Coimbatore', 'BANK000002');

INSERT INTO Branch VALUES
(3, 'Madurai Branch', 'Madurai', 'BANK000003');

INSERT INTO Branch VALUES
(4, 'Trichy Branch', 'Trichy', 'BANK000004');

INSERT INTO Customer VALUES
(1, 'Deepika', DATE '2004-05-15', 'Female',
'9876543210', 'deepika@gmail.com', 'Chennai');

INSERT INTO Customer VALUES
(2, 'Priya', DATE '2003-08-20', 'Female',
'9876543211', 'priya@gmail.com', 'Madurai');

INSERT INTO Customer VALUES
(3, 'Arun', DATE '1998-02-10', 'Male',
'9876543212', 'arun@gmail.com', 'Coimbatore');

INSERT INTO Customer VALUES
(4, 'Karthik', DATE '1995-11-25', 'Male',
'9876543213', 'karthik@gmail.com', 'Trichy');

INSERT INTO Customer VALUES
(5, 'Meena', DATE '1997-07-12', 'Female',
'9876543214', 'meena@gmail.com', 'Chennai');

INSERT INTO Account VALUES
(100001, 1, 1, 'Savings', DATE '2024-01-10',
50000.00, 'Active');

INSERT INTO Account VALUES
(100002, 2, 3, 'Savings', DATE '2023-06-15',
75000.00, 'Active');

INSERT INTO Account VALUES
(100003, 3, 2, 'Current', DATE '2022-03-20',
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150000.00, 'Active');

INSERT INTO Account VALUES
(100004, 4, 4, 'Savings', DATE '2024-08-05',
30000.00, 'Active');

INSERT INTO Account VALUES
(100005, 5, 1, 'Savings', DATE '2025-01-15',
90000.00, 'Active');

INSERT INTO Bank_Transaction VALUES
(5001, 100001, DATE '2026-09-01',
'Deposit', 10000.00, 'Cash deposit');

INSERT INTO Bank_Transaction VALUES
(5002, 100001, DATE '2026-09-05',
'Withdrawal', 5000.00, 'ATM withdrawal');

INSERT INTO Bank_Transaction VALUES
(5003, 100002, DATE '2026-09-02',
'Deposit', 20000.00, 'Salary credit');

INSERT INTO Bank_Transaction VALUES
(5004, 100003, DATE '2026-09-03',
'Withdrawal', 15000.00, 'Business payment');

INSERT INTO Bank_Transaction VALUES
(5005, 100004, DATE '2026-09-04',
'Deposit', 5000.00, 'Cash deposit');

INSERT INTO Bank_Transaction VALUES
(5006, 100005, DATE '2026-09-05',
'Deposit', 25000.00, 'Salary credit');

INSERT INTO Loan VALUES
(7001, 1, 'Education Loan', 200000.00,
7.50, DATE '2025-06-10', 'Active');

INSERT INTO Loan VALUES
(7002, 2, 'Home Loan', 1500000.00,
```

```
8.25, DATE '2024-04-15', 'Active');

INSERT INTO Loan VALUES
(7003, 3, 'Business Loan', 500000.00,
9.00, DATE '2023-08-20', 'Active');

INSERT INTO Loan VALUES
(7004, 4, 'Personal Loan', 300000.00,
10.50, DATE '2025-01-20', 'Closed');

INSERT INTO Loan_Payment VALUES
(8001, 7001, DATE '2026-08-10',
10000.00, 'UPI');

INSERT INTO Loan_Payment VALUES
(8002, 7002, DATE '2026-08-15',
25000.00, 'Bank Transfer');

INSERT INTO Loan_Payment VALUES
(8003, 7003, DATE '2026-08-20',
20000.00, 'Cheque');

INSERT INTO Loan_Payment VALUES
(8004, 7004, DATE '2026-08-25',
30000.00, 'UPI');

COMMIT;

SELECT * FROM Customer;

/*
OUTPUT:

CUSTOMER_ID CUSTOMER_NAME DATE_OF_BIRTH GENDER
1           Deepika       15-MAY-04     Female
2           Priya         20-AUG-03     Female
3           Arun          10-FEB-98     Male
4           Karthik       25-NOV-95     Male
5           Meena         12-JUL-97     Female
```

PHONE EMAIL ADDRESS
9876543210 deepika@gmail.com Chennai
9876543211 priya@gmail.com Madurai
9876543212 arun@gmail.com Coimbatore
9876543213 karthik@gmail.com Trichy
9876543214 meena@gmail.com Chennai
*/

SELECT * FROM Branch;

/*
OUTPUT:

BRANCH_ID BRANCH_NAME CITY IFSC_CODE
1 Chennai Main Branch Chennai BANK000001
2 Coimbatore Branch Coimbatore BANK000002
3 Madurai Branch Madurai BANK000003
4 Trichy Branch Trichy BANK000004
*/

SELECT * FROM Account;

/*
OUTPUT:

ACCOUNT_NO CUSTOMER_ID BRANCH_ID ACCOUNT_TYPE
100001 1 1 Savings
100002 2 3 Savings
100003 3 2 Current
100004 4 4 Savings
100005 5 1 Savings

OPENING_DATE BALANCE ACCOUNT_STATUS
10-JAN-24 50000 Active
15-JUN-23 75000 Active
20-MAR-22 150000 Active
05-AUG-24 30000 Active
15-JAN-25 90000 Active
*/

```
SELECT
   c.Customer_ID,
   c.Customer_Name,
   a.Account_No,
   a.Account_Type,
   a.Balance,
   a.Account_Status
FROM Customer c
JOIN Account a
ON c.Customer_ID = a.Customer_ID;

/*
OUTPUT:

CUSTOMER_ID CUSTOMER_NAME ACCOUNT_NO ACCOUNT_TYPE BALANCE
STATUS
1           Deepika       100001     Savings      50000    Active
2           Priya         100002     Savings      75000    Active
3           Arun          100003     Current      150000   Active
4           Karthik       100004     Savings      30000    Active
5           Meena         100005     Savings      90000    Active
*/

SELECT
   c.Customer_Name,
   a.Account_No,
   b.Branch_Name,
   b.City,
   b.IFSC_Code
FROM Customer c
JOIN Account a
ON c.Customer_ID = a.Customer_ID
JOIN Branch b
ON a.Branch_ID = b.Branch_ID;

/*
OUTPUT:

CUSTOMER_NAME ACCOUNT_NO BRANCH_NAME          CITY        IFSC_CODE
Deepika       100001     Chennai Main Branch  Chennai     BANK000001
```

```
Priya        100002    Madurai Branch       Madurai     BANK000003
Arun         100003    Coimbatore Branch    Coimbatore  BANK000002
Karthik      100004    Trichy Branch        Trichy      BANK000004
Meena        100005    Chennai Main Branch  Chennai     BANK000001
*/

SELECT
   Account_No,
   Customer_ID,
   Balance
FROM Account
WHERE Account_Type = 'Savings';

/*
OUTPUT:

ACCOUNT_NO CUSTOMER_ID BALANCE
100001     1           50000
100002     2           75000
100004     4           30000
100005     5           90000
*/

SELECT
   Account_No,
   Customer_ID,
   Account_Type,
   Balance
FROM Account
WHERE Balance > 50000;

/*
OUTPUT:

ACCOUNT_NO CUSTOMER_ID ACCOUNT_TYPE BALANCE
100002     2           Savings      75000
100003     3           Current      150000
100005     5           Savings      90000
*/
```

```
SELECT * FROM Bank_Transaction;

/*
OUTPUT:

TRANSACTION_ID ACCOUNT_NO TRANSACTION_DATE TRANSACTION_TYPE
AMOUNT
5001        100001    01-SEP-26      Deposit        10000
5002        100001    05-SEP-26      Withdrawal      5000
5003        100002    02-SEP-26      Deposit        20000
5004        100003    03-SEP-26      Withdrawal      15000
5005        100004    04-SEP-26      Deposit        5000
5006        100005    05-SEP-26      Deposit        25000

DESCRIPTION
Cash deposit
ATM withdrawal
Salary credit
Business payment
Cash deposit
Salary credit
*/

SELECT
   c.Customer_Name,
   a.Account_No,
   t.Transaction_Date,
   t.Transaction_Type,
   t.Amount,
   t.Description
FROM Bank_Transaction t
JOIN Account a
ON t.Account_No = a.Account_No
JOIN Customer c
ON a.Customer_ID = c.Customer_ID;

/*
OUTPUT:

CUSTOMER_NAME ACCOUNT_NO TRANSACTION_DATE TYPE       AMOUNT
```

```
Deepika       100001     01-SEP-26        Deposit    10000
Deepika       100001     05-SEP-26        Withdrawal 5000
Priya         100002     02-SEP-26        Deposit    20000
Arun          100003     03-SEP-26        Withdrawal 15000
Karthik       100004     04-SEP-26        Deposit    5000
Meena         100005     05-SEP-26        Deposit    25000
*/

SELECT
   SUM(Amount) AS Total_Deposits
FROM Bank_Transaction
WHERE Transaction_Type = 'Deposit';

/*
OUTPUT:

TOTAL_DEPOSITS
60000
*/

SELECT
   SUM(Amount) AS Total_Withdrawals
FROM Bank_Transaction
WHERE Transaction_Type = 'Withdrawal';

/*
OUTPUT:

TOTAL_WITHDRAWALS
20000
*/

SELECT
   Account_No,
   Customer_ID,
   Balance
FROM Account
WHERE Balance = (SELECT MAX(Balance) FROM Account);

/*
```

```
OUTPUT:

ACCOUNT_NO CUSTOMER_ID BALANCE
100003     3           150000
*/

SELECT
   Account_No,
   Customer_ID,
   Balance
FROM Account
WHERE Balance = (SELECT MIN(Balance) FROM Account);

/*
OUTPUT:

ACCOUNT_NO CUSTOMER_ID BALANCE
100004     4           30000
*/

SELECT
   Account_Type,
   COUNT(*) AS Number_Of_Accounts
FROM Account
GROUP BY Account_Type;

/*
OUTPUT:

ACCOUNT_TYPE NUMBER_OF_ACCOUNTS
Savings      4
Current      1
*/

SELECT
   AVG(Balance) AS Average_Balance
FROM Account;

/*
OUTPUT:
```

```
AVERAGE_BALANCE
79000
*/

SELECT * FROM Loan;

/*
OUTPUT:

LOAN_ID CUSTOMER_ID LOAN_TYPE       LOAN_AMOUNT INTEREST_RATE
LOAN_DATE
7001    1           Education Loan  200000      7.5           10-JUN-25
7002    2           Home Loan       1500000     8.25          15-APR-24
7003    3           Business Loan   500000      9             20-AUG-23
7004    4           Personal Loan   300000      10.5          20-JAN-25

LOAN_STATUS
Active
Active
Active
Closed
*/

SELECT
   Loan_ID,
   Customer_ID,
   Loan_Type,
   Loan_Amount,
   Interest_Rate,
   Loan_Status
FROM Loan
WHERE Loan_Status = 'Active';

/*
OUTPUT:

LOAN_ID CUSTOMER_ID LOAN_TYPE       LOAN_AMOUNT RATE STATUS
7001    1           Education Loan  200000      7.5  Active
7002    2           Home Loan       1500000     8.25 Active
```

```
7003    3          Business Loan   500000      9    Active
*/

SELECT
   c.Customer_Name,
   l.Loan_ID,
   l.Loan_Type,
   l.Loan_Amount,
   l.Interest_Rate,
   l.Loan_Status
FROM Customer c
JOIN Loan l
ON c.Customer_ID = l.Customer_ID;

/*
OUTPUT:

CUSTOMER_NAME LOAN_ID LOAN_TYPE       LOAN_AMOUNT RATE STATUS
Deepika       7001    Education Loan  200000      7.5  Active
Priya         7002    Home Loan       1500000     8.25 Active
Arun          7003    Business Loan   500000      9    Active
Karthik       7004    Personal Loan   300000      10.5 Closed
*/

SELECT
   SUM(Loan_Amount) AS Total_Loan_Amount
FROM Loan;

/*
OUTPUT:

TOTAL_LOAN_AMOUNT
2500000
*/

SELECT
   Loan_ID,
   Customer_ID,
   Loan_Type,
   Loan_Amount
```

```
FROM Loan
WHERE Loan_Amount = (SELECT MAX(Loan_Amount) FROM Loan);

/*
OUTPUT:

LOAN_ID CUSTOMER_ID LOAN_TYPE  LOAN_AMOUNT
7002    2           Home Loan  1500000
*/

SELECT * FROM Loan_Payment;

/*
OUTPUT:

PAYMENT_ID LOAN_ID PAYMENT_DATE PAYMENT_AMOUNT PAYMENT_MODE
8001       7001    10-AUG-26   10000          UPI
8002       7002    15-AUG-26   25000          Bank Transfer
8003       7003    20-AUG-26   20000          Cheque
8004       7004    25-AUG-26   30000          UPI
*/

SELECT
   c.Customer_Name,
   l.Loan_Type,
   lp.Payment_Date,
   lp.Payment_Amount,
   lp.Payment_Mode
FROM Loan_Payment lp
JOIN Loan l
ON lp.Loan_ID = l.Loan_ID
JOIN Customer c
ON l.Customer_ID = c.Customer_ID;

/*
OUTPUT:

CUSTOMER_NAME LOAN_TYPE       PAYMENT_DATE PAYMENT_AMOUNT MODE
Deepika       Education Loan  10-AUG-26    10000          UPI
Priya         Home Loan       15-AUG-26    25000          Bank Transfer
```

```
Arun          Business Loan   20-AUG-26    20000        Cheque
Karthik       Personal Loan   25-AUG-26    30000        UPI
*/

SELECT
   SUM(Payment_Amount) AS Total_Loan_Payments
FROM Loan_Payment;

/*
OUTPUT:

TOTAL_LOAN_PAYMENTS
85000
*/

UPDATE Account
SET Balance = Balance + 5000
WHERE Account_No = 100001;

/*
OUTPUT:

1 row updated.

Account 100001 balance:
55000
*/

UPDATE Customer
SET Phone = '9999999999'
WHERE Customer_ID = 1;

/*
OUTPUT:

1 row updated.

Deepika's new phone number:
9999999999
*/
```

```
UPDATE Loan
SET Loan_Status = 'Closed'
WHERE Loan_ID = 7004;

/*
OUTPUT:

1 row updated.

Loan 7004 status:
Closed
*/

DELETE FROM Loan_Payment
WHERE Payment_ID = 8004;

/*
OUTPUT:

1 row deleted.

Payment 8004 has been deleted.
*/

COMMIT;

/*
OUTPUT:

Commit complete.
*/

SELECT
   Customer_ID,
   Customer_Name,
   Phone
FROM Customer;

/*
```

OUTPUT:

CUSTOMER_ID CUSTOMER_NAME PHONE
1 Deepika 9999999999
2 Priya 9876543211
3 Arun 9876543212
4 Karthik 9876543213
5 Meena 9876543214
*/

```
SELECT
   Account_No,
   Balance
FROM Account;
```

/*
OUTPUT:

ACCOUNT_NO BALANCE
100001 55000
100002 75000
100003 150000
100004 30000
100005 90000
*/

```
SELECT
   Payment_ID,
   Loan_ID,
   Payment_Amount
FROM Loan_Payment;
```

/*
OUTPUT:

PAYMENT_ID LOAN_ID PAYMENT_AMOUNT
8001 7001 10000
8002 7002 25000
8003 7003 20000
*/